

Malcolm Lelis Afonso

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I am seeking an opportunity to provide my expertise to the masses in order to help them make informed business decisions as a Drone (UAV) Remote Sensing and GIS Expert.

ACADEMIC QUALIFICATION

Program	Institution	%	Completion
Master of Science	Goa University Taleigao Plateau, Goa University	GPA 7.78	2015
Bachelor of Science	SMT. Parvatibai Chowgule College of Arts and Science Margao, Goa.	76%	2013
Diploma in Nautical Science	NAMAC(IMS Vikrant) Mumbai, IGNOU	78%	2009
XII	Fr.Agnel HSSC Verna, Goa Board	59 %	2008
X	Loyola HS Margao, Goa board	69 %	2006

TECHNICAL SKILLS

- MS-OFFICE
- Remote Sensing and GIS (QGIS, Google Earth Engine, SAGA, GPS, Post-GIS)
- Basic Programming (Python, SQL)
- DGCA certified Drone Pilot- <https://digitalsky.dgca.gov.in/rpc?pilotId=9395ba6c-d1d0-4a65-a28c-7947c46f7ed0>
- Drone Photogrammetry

WORK EXPERIENCE

- **Founder, Afonso Spatialcraft LLP** **Feb'24-Present**
As the founder and leader of Spatialcraft, a cutting-edge geospatial technology company, I spearhead initiatives that deliver innovative GIS solutions, focusing on Smart Interactive Digital Maps (SIDM) for real estate, government services, and infrastructure development. I work closely with DGPS survey teams to ensure precise data collection and project execution. Additionally, I integrate CAD and GIS data across platforms such as QGIS and AutoCAD, enabling seamless cross-platform functionality. Notably, I led the geotagging and mapping project for Smart City Panjim, Goa, with a focus on tree geotagging and urban planning. Under my leadership, Spatialcraft was recognized with the 'Best Innovation in Drone Technology' award at a national startup conclave, underscoring our contributions to the geospatial industry www.spatialcraft.in
- **GIS Consultant, Goa Electronics Limited, Panaji Goa.** **April 23-Present**
As a GIS Consultant at Goa Electronics Limited (GEL) since April 2023, I have been instrumental in the development of the OneMapGoa GIS platform, a decision support system integrating geospatial data for government departments and citizens across Goa. My role involves creating the state basemap and integrating utilities data from various departments such as Electricity, PWD, and WRD, utilizing advanced data cleaning, editing, and visualization techniques. I guide the team in preparing scope documents, designing technical workflows, and setting GIS standards to ensure seamless data integration. Additionally, I collaborate on optimizing system architecture, enhancing frontend interfaces, and incorporating open-source tools for efficient geospatial processing. By evaluating departmental needs, establishing data management protocols, and conducting training workshops for officials, I contribute to the platform's effectiveness in advancing e-governance, planning, and resource management across the state.

- Founder, Geosensitive- Remote sensing and GIS services.** **July 19- Present**
 As the founder of GeoSensitive, I lead the delivery of advanced GIS and Drone (UAV) Remote Sensing Services. We successfully submitted IBM compliance reports for mines in Goa and Andhra Pradesh, meeting stringent regulatory standards. Additionally, we completed various projects mapping agricultural land in Goa's villages, providing valuable geospatial insights for local development.
- Mapping of Peoples Draft Coastal Zone Management Plans for Villages across Goa** **Aug'19-Sept'19**
 As a part of the community outreach initiative, assisted local communities in preparing their Draft Coastal Zone Management Plans including mapping of important coastal features such as ShellFish Breeding Grounds, Important Fishing Zones, Canoe ramps, Ferry ramps, Fishermen Huts and other community structures such as Bunds, Sluice Gates, Ponds by associating with a local NGO along with the assistance of the representatives of the concerned communities such as farmers, fishermen, toddy tappers etc.
- Assistant Professor at Parvatibai Chowgule College of Arts and Science (Autonomous)** **Jan'16-Aug'22**
 My work involved conducting lectures and practicals, developing innovative teaching-learning-evaluation methodologies using various information and communication technology tools, in particular, I adopted a flipped-classroom approach. I curated and launched several new courses in the department including "Remote sensing and digital image processing" and "GIS for Geoscientist" I also took charge of TyBsc research projects. Conducted field visits to research institutions of repute, organized field training for students in various parts of Goa, organized industry-student interaction seminars, Student skill enhancement workshops and was actively involved in coaching college football team for one year. I initiated a geology departmental blog, e-newsletter, alumni association of the department, community outreach activities etc. I am a regular member of the staff football and cricket team. I was also assigned as the member secretary of the geology department board of studies.
- Geologist at Soiltech India Private Limited** **Jun'15-Nov'15**
 - As a site in charge of **Beach sand and offshore investigation** project (**sonic drilling**) in Kerala for Atomic minerals division, Govt of India; My work involved onsite project management including, working with multinational drilling crew, Coordinating with clients, taking locals into confidence about the proposed borehole location, Preparing various site reports (**daily drilling report, account of expenditure, fuel purchase, site requirements**), responsible for the overall execution of the project in a safe and efficient manner.
 - As a Site in charge of **Geotechnical Investigation** of a Flyover project in Ponda Goa, my work involved carrying out **Standard penetration tests**, soil and rock **sampling**, preparing **Borehole Log, Reporting** to clients and company, **Coordinating** with water suppliers, besides overlooking the activities of seven drilling crew personnel and the working of two calyx rigs.

INTERNSHIPS

- Summer training programme at ONGC Cambay sub-asset** **19'May'14-18'June'14**
 Attended one month of internship programme offered by ONGC, Cambay sub-asset. Programme involved interactive sessions with the experts from the different departments at the base followed by field visits to Oil rigs, Group gathering stations (GGS), Sucker rod pump (SRP).
- Trainee Navigating Officer, MT Torm Fox Vessel** **Sept'09-Apr'10**
 I was responsible for cargo paperwork, ship provisions besides discharging regular navigation and cargo watchkeeping duties.

ACADEMIC PROJECTS

- **QGIS Training for the technical staff of the Directorate of Settlement and Land Records, Panaji-Goa** Jan'22
conducted a Six-Day Short Term Training Program on QGIS, a Curated programme for the technical staff of the Directorate of Settlement and Land Records, Panaji-Goa from 31-1-2022 to 5-2-2022. The course was attended by 9 technical staff of the Directorate of Settlement and Land Records.
- **Study of Landslide Affected Areas in Ponda, Goa.** Jun'18-Mar'19
In this study an attempt was made to understand the underlying causes for landslides, both natural as well as anthropogenic factors were identified by carrying out an extensive field study of the locality and mitigation measures were suggested to reduce the occurrence of landslides.
- **Micro Level Study of Landslide Affected Areas in Sada, Vasco, Goa.** Jan'17-Mar'18
In this study, an attempt was made to understand the underlying causes for landslides, both natural as well as anthropogenic factors were identified by carrying out an extensive field study of the locality and mitigation measures were suggested to reduce the occurrence of landslides.
- **Study of The Groundwater Regime, Its Utilization and Quality in Betul Village, Salcete Goa** Jan'16-Mar'17
Guided a group of four students in their final year research project. Pre-monsoon and post-monsoon water samples were collected from wells located in the village and their chemical parameters were analyzed. The results were then compared to standard drinking water permissible limits.
- **Study of laterite weathering and profile development over mafic rock basement.** Sept'14-April'15
Studied the laterite profile formed over metabasalt, prepared a section of the profile demarcating and designation names to the various reaction zones, and described the mineralogy (XRD analysis), textural and structural attributes of the different zones.
- **Petrology and structural features of basement rocks of Kaladgi supergroup at and around Ramthal, Bagalkot district, Karnataka.** Sept'12-Mar'13
In a team of 4, studied the outcrop of rocks (dominantly Banded Hematite Quartzite) exposed at Ramthal for their structural and chemical characteristics. The structural features such as folding, faults, and warping were related to the three major regional events of deformation noted for this region in the previous literature and also iron ore chemical analysis was done for these rocks that yielded low iron content

TRAINING & SEMINARS

- Successfully completed ten days faculty development program on **Artificial Intelligence** organized by Panimalar Engineering College, Chennai and Foursteps training solutions in Online mode. 2-11'July'21
- Participated in the "**Webinar Training on Flood Mapping using SAR**" on TGIS Online Platform, Ahmedabad(Gujarat). 26'June'21
- Participated in the "**Webinar Training on Surface Runoff Modelling (QGIS)**" on TGIS Online Platform, Ahmedabad(Gujarat). 12'June'21
- Successfully completed the "**How to write a successful research paper**" online course by Andre Klapper, Ph.D., Instructor & Udemy. 21'April'21
- Successfully completed the webinar "**Hyperspectral Data for Land and Coastal Systems**" organized by NASA's Applied Remote Sensing Training Program. 19,26,'Jan-2'Feb'21
- Successfully participated in a two-day workshop on "**Research Grant Writing**" organized by the Directorate of Higher Education, Goa in association with the Government College of Arts, Science & Commerce, Quepem Goa at SCERT Building, Porvorim-Goa. 16-17'Mar'21
- Successfully completed the webinar "**Mapping and Monitoring Lakes and Reservoirs with Satellite Observations**".organized by NASA's Applied Remote Sensing Training Program 9,16,23'Feb'21

- Successfully completed 5 total hours of **"Statistics for Data Science and Business Analysis"** online course offered by 365 careers and Udemy.
25'Jan'21
- Successfully completed advance module (Track 2) of the MOOC on **"Geospatial Applications for Disaster Risk Management"** jointly conducted by the Centre for Space Science and Technology Education for Asia and the Pacific (CSSTEAP) & United Nations Platform for Space based Information for Disaster Management and Emergency Response (UN-SPIDER).
Oct-Nov'2020
- Attended short term Web-based training course on **"Satellite based Flood Assessment and Modeling"** under TREES (Training and Research in Earth Eco-System), an initiative of space Applications Centre (ISRO), Ahmedabad during 03-05 November, 2020.
- **Copernicus MOOC**, University of Luxembourg Competence Centre.
Mar-May'20
- Participated in a regional workshop on **Research Based Pedagogical Tools** held at Parvatibai Chowgule College of Arts and Science (Autonomous), Gogol-Margao, Goa. The workshop was organized by Centre of Excellence in Science and Mathematics Education, IISER Pune.
5-7'Mar'20
- **Learn to code in Python 3:** Programming beginner to advance, Ivan Lourenci Gomes and Udemy.
Mar'20
- **DRONE01x: Drones for Agriculture:** Prepare and Design Your Drone (UAV) Mission, WageningenX, an online learning initiative of Wageningen University & Research.
Jan' 20
- Participated in 5 days' **workshop in Petrology** organized by the Department of Geology, Parvatibai Chowgule College of Arts and Science-Autonomous, Margao, Goa.
4-8'Dec'18
- Successfully Completed the **Field Training for Graduate/Post Graduate Geology Teachers** at FTC, Chitradurga Conducted by the Geological Survey of India Training Institute, Hyderabad.
10-16'Jun'18
- Participated in The Two Day State Level Workshop on **Flipped Learning** Organized by IQAC, Parvatibai Chowgule College of Arts and Science-Autonomous, Margao, Goa.
8-9'Nov'17
- Successfully Completed The 4 Weeks Online Course Titled **"Digital Image Processing of Remote Sensing Data"** With A Score Of 70% at NPTEL Online Certification, Ministry of HRD, Govt. of India.
Jun'17-Aug'17
- Attended 21 days **Summer Training on Geospatial technologies**, under Natural Resources Data Management System(NRDMS) programme of DST, Ministry of Science and Technology, Government of India, Conducted by the Department of Remote Sensing and Technology, Birla Institute of Scientific Research, Jaipur.
15'May'17-5'June'17
- Successfully completed the course **"ICT Integration In Higher Education"** Offered by SNDT Women's University.
Oct'17-Dec'17
- Participated in the two-day state level Train the Trainer workshop on **MOOCs-Design, Development & Delivery** organised by IQAC, Parvatibai Chowgule College of Arts and Science
Oct'16
- Attended Certificate course on **Basics in GIS Technology** for teachers and researchers organized by Department of Geography & Research Centre, Parvatibai Chowgule College, Margao, Goa.
April'16
- Completed the basic course on **'Remote sensing, Geographical Information system and Global Navigation Satellite System'** conducted by Indian Institute Of Remote Sensing.
Aug'14-Nov'14
- Attended seminar on **'Recent trends in management and conservation of groundwater resources'**
Feb'13
- Chowgule College Accredited Honours Programme entitled **"Basics of GIS & GPS"**
2011-2012

ACHIEVEMENTS

- **Best Innovation in Drone Technology** award at **Startup ka Mahakumbh**, India's first startup conclave and expo, held at the India International Convention Centre, Yashobhoomi, New Delhi.
Jun'24
- Goa University **Geology** Department **Gold** medallist, batch of 2015.
Jun'15
- Won **1st** place in **Geonny Quest** at TERRA organised by St.Xaviers Institute (Autonomous) Mumbai
Dec'13
- Awarded **Letter of recognition** for organising a state-level seminar on 'Recent trends in management and conservation of groundwater resources' organized by the Department of Geology in association with the Department of Science Technology and Environment, Govt of Goa.
Feb'13
- Recipient of **Gold** medal as a regular member of college sports team in inter-collegiate **football, handball** and **hockey** tournaments organized by Goa University in 2012 and 2013".

EXTRA-CURRICULAR ACTIVITIES

- **Betalbatim Village Farmers Club:** Currently I am assisting in the revival of the agricultural sector in Betalbatim village as a member secretary of the village farmers club. **Aug'20**
- **Organized a student seminar** under DBT Star College Scheme, on the topic "Coastal Regulation Zone Management in Goa: CRZ and CZMP" **06'Aug'19**
- **Organized 2 day's hands on workshop** on Remote Sensing and GIS in association with Geotech GIS institute and Consultancy, Aurangabad, Maharashtra. **22-23'Feb'19**
- **Organized seminar** on the topics 1) insights on Mining in Goa and 2) Mine land reclamation in association with Sesa Goa Iron Ore a Vedanta company. **Feb'19**
- **Nobel Prize Series:** Volunteered as a member of the organizing committee for the Nobel Prize Series, India 2018 event organized by Media AB, Department of Biotechnology, Govt. of India and Department of Science and Technology, Govt. of Goa. **Feb'18**
- **Departmental newsletter:** Was in charge of Departmental newsletter Geos Vol VII (I). **Sept'17**
- **Workshop on GIS:** Conducted Two day workshop titled "Learn to create your first map" using QGIS and Google Earth. **Jan'18**
- **Departmental Blog:** Started the Departmental Blog. <https://geology-chowgules.blogspot.in/>
- **Departmental Alumni Association:** Initiated the process of formation of the Departmental Alumni Association.
- **Geological Society of Goa:** Appointed as a member of the executive committee of the Geological Society of Goa. Volunteered in organizing society events namely annual trek and release of the book titled Geology and Mineral Resources of Goa authored by Ashoka G Dessai.
- **Coordinator** for Geology Department for organizing Nebula, annual geology departmental event with participation of about seven colleges/institutes from Maharashtra, Karnataka and Goa. **2013**
- Member of **Sports Council** of Smt. Parvatibai Chowgule College **Jun'12-Mar'13**
- **Secretary** of Tiger Football Club affiliated to Smt. Parvatibai Chowgule College with a membership of over **1000** students and alumni, founder of Joga Bonito Futsal League and TFC pro Tourney, was responsible for the smooth conduct of the Annual Weekly Training Programme (**AWTP**) including handling of accounts, coach appointments, infrastructure bookings, maintaining attendance registers, and also handling of equipment. **Jun'11-Mar'12**
- Member of **Students council** of Smt. Parvatibai Chowgule College **Jun'10-Mar'11**